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PRACTICAL WEEK 3

Task 1: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process States at different stages of execution

CODE:

```
UW PID0 6.09 File: lk.c

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("=== Process Creation using fork() ===\n");
    printf("Before fork()\n");
    printf("PID : %d\n", getpid());
    printf("PPID : %d\n", getppid());
    printf("State: Running\n");

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed.\n");
        return 1;
    }

    if (pid == 0)
    {
        /* Child Process */
        printf("=== Child Process ===\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
        printf("State: Running\n");

        sleep(2);

        printf("\nChild Process:\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
        printf("State: Exiting\n");

        exit(0);
    }
    else
    {
        /* Parent Process */
        printf("=== Parent Process ===\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
        printf("Child PID: %d\n", pid);
        printf("State: Running\n");

        printf("\nParent is waiting for child...\n");
        printf("State: Waiting\n");

        wait(NULL);

        printf("\nChild process has terminated.\n");
        printf("Parent Process:\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
    }
}
```

OUTPUT:

```
nrk@nallas-mac-Book-Air ~ % gcc lk.c -o lk
nrk@nallas-mac-Book-Air ~ % ./lk
=== Process Creation using fork() ===
Before fork():
PID : 1541
PPID : 1541
State: Running

=== Parent Process ===
PID : 1541
PPID : 1541
Child PID: 1542
State: Running

Parent is waiting for child...
State: Waiting
=== Child Process ===
PID : 1542
PPID : 1541
State: Running

Child Process:
PID : 1542
PPID : 1541
State: Exiting

Child process has terminated.
Parent Process:
PID : 1541
PPID : 1541
State: Running

Parent Process State: Exiting
nrk@nallas-mac-Book-Air ~ %
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Code:

```
uw PIDO 6.09 File: 13.c

#include <stdio.h>
#include <stdlib.h>

int main()
{
    printf("=====\\n");
    printf("  HARDWARE AND OS RESOURCE ANALYSIS\\n");
    printf("=====\\n\\n");

    printf("1. OPERATING SYSTEM INFORMATION\\n");
    printf("-----\\n");
    system("uname -a");

    printf("2. CPU INFORMATION\\n");
    printf("-----\\n");
    system("lscpu");

    printf("3. RUNNING PROCESSES\\n");
    printf("-----\\n");
    system("ps aux");

    printf("4. REAL-TIME SYSTEM RESOURCE USAGE\\n");
    printf("-----\\n");
    printf("Press Ctrl+C to exit the top command.\\n\\n");
    system("top");

    return 0;
}
```

Output:

Processes: 633 total, 2 running, 631 sleeping, 3217 threads																											08:37:09	
Load Avg: 3.68, 110.26, 145.48 CPU usage: 4.43% user, 5.90% sys, 89.63% idle. Sharedlib: 358M resident, 80M data, 64M linkedin. MemRegions: 0 total, 88 resident, 185M private, 787M shared.																												
PhysMem: 7444M used (1024M wired, 7837M compressed), 140M unused, 488(0) Mapped, 484(4) M Inactive, 575(12) M Free.																												
Disks: 1186378/28G read, 688999/9085M written.																												
PID	COMMAND	NCPU	TIME	PTH	#WG	#PORT	MEM	PURG	CMRPS	PGRP	PPID	STATE	BOOSTS	NCPU	MEM	NCPU_OTHERS	UID	FAULTS	COW	MSGSENT	MSGRCV	SYSSBD	SYMSZ	CSW	PAGEIN	IDLEW	POWE	
0	kernel_task	18.0	04:16:18.555	0	0	0	17M	08	08	0	0	running	[0]	0.00000	0.00000	0	34781	2	3656444	3154884	0	0	8135423	0	85178	0.0		
1845	Comet Helper	18.0	04:16:17.74	21	3	227	110M	08	41M	965	965	sleeping	*[22117]	1.17000	0.00000	501	79634	933	450804	195434	433618	1742794	518584	1517	474	18.0		
965	Comet	0.4	01:18:59.41	1	692	131M	08	90M	965	1	sleeping	*[28315]	0.00000	0.00000	501	131754	1211	457493	227179	456794	2488661	198164	17252	1938	8.5			
487	WindowServer	6.3	03:56:79.20	5	4457	694M	21M	231M	487	1	sleeping	[0]	0.07162	1.48336	88	318348	18429	2193377	1747853	2498327	5442262	1892749	5556	2286	5.4			
875	Google Chrom	4.7	03:25:08.19	1	141	41M	08	23M	441	441	sleeping	[0]	0.00000	0.00000	501	68564	833	39083	16419	1785483	30954	399769	4497	4	4.7			
1679	top	4.3	00:01:08.1/1	0	29	7984K	08	88	1675	1675	running	[0]	0.00000	0.00000	0	74364	95	988521	498272	36294	541654	464	29	0	4.3			
441	mDNSResponder	4.2	00:18:97.4	2	132	9898K	08	1688K	441	1	sleeping	[0]	0.00000	0.00000	65	14379	154	8073	4434	843245	148674	192288	566	1845	4.4			
558	com.apple.dt	2.7	00:13:04.08	6	1346	30M	08	4192K	558	1	sleeping	[0]	0.00000	0.00000	278	28834	63	74194	57716	298744	611488	488363	1191	43181	10.1			
874	Google Chrom	2.3	00:41:31.24	5	385	350M	08	48M	441	441	sleeping	[10]	0.05633	0.00000	501	99436	926	383233	135948	294889	889367	347713	3644	287	2.3			
626	trustd	1.8	00:19:28.3	2	259	9648K	512K	3748K	626	1	sleeping	[0]	0.00000	1.78015	501	28993	122	4513	4186	118757	38635	51577	91	1	1.8			
441	Google Chrom	1.7	00:14:48.59	2	958	160M	08	37M	441	1	sleeping	[0]	0.00000	0.00000	501	224688	1323	356614	156437	481568	1198841	43739	27765	62	1.7			
644	WhatsApp	1.6	01:48:41.26	4	688	287M	1024K	135M	644	1	sleeping	[10]	0.18765	0.00000	501	329333	3292	76418	26911	1541529	628433	684488	36979	89	1.6			
399	bluetoothd	1.4	00:18:59.12	6	480	18M	256K	3648K	399	1	sleeping	[0]	1.57497	0.00000	0	25736	184	17639	7138	164729	31355	73865	4543	3	1.4			
421	nautilus	1.4	00:03:25.8	5	178	8592K	08	1448K	421	1	sleeping	[0]	2.08168	0.00000	501	14513	152	5454	2880	68381	11769	22238	376	0	1.4			
369	coreduetd	1.2	00:03:19.8	7	98	7249K	384K	1778K	369	1	sleeping	*[149]	0.00000	1.16146	0	28888	158	4626	2348	182472	19378	3685	1286	0	1.2			
753	corespotlight	1.1	00:16:76.8	6	382	25M	08	14M	753	1	sleeping	[0]	3.02888	0.00000	501	141911	381	48392	26874	517111	59183	282434	18318	1	1.1			
693	suggestd	0.9	00:07:76.7	6	254	13M	08	3712K	693	1	sleeping	[0]	0.00000	0.08467	501	28713	243	11841	2619	73986	25498	57258	3390	1	0.9			
614	cloudot	0.7	00:18:55.7	4	297	16M	08	4848K	614	1	sleeping	[0]	0.00000	0.28873	501	42331	286	47845	21883	326218	85836	147206	4551	11	0.7			
978	helpd	0.7	00:18:39.8	6	93	17M	08	8112K	978	1	sleeping	[0]	0.00000	0.16449	501	6592	153	1338	318	44898	3725	12675	993	3	0.7			
788	aid	0.5	00:01:04.2	1	36	2336K	08	488K	788	1	sleeping	[0]	0.00000	0.09224	265	7268	181	3176	2651	25778	11523	6525	883	0	0.5			
489	cfprefd	0.5	00:05:25.3	2	952	3856K	44K	1184K	489	1	sleeping	[0]	0.00000	0.05765	0	18482	98	16584	16547	208119	42144	48474	14	0	0.5			
657	Terminal	0.5	00:09:08.9	4	343	151M	21M	14M	657	1	sleeping	[0]	0.03113	0.03966	501	73791	411	85543	12656	138892	288145	77348	6365	78	0.5			
653	sharings	0.4	00:08:47.4	2	348	17M	08	8784K	653	1	sleeping	[0]	0.00000	0.13564	501	28686	291	28661	9211	98824	62646	74174	3155	1	0.4			
789	Finder	0.4	00:25:28.7	4	549	161M	08	135M	789	1	sleeping	[23]	0.04518	0.00000	501	157778	542	139782	58225	186248	356963	155165	7435	115	0.4			
568	searchpartyd	0.4	00:18:39.5	4	98	8359K	08	3824K	568	1	sleeping	[0]	0.00000	0.32918	0	11258	156	1855	1551	39188	4429	18889	2483	0	0.4			
681	afirm2m	0.4	00:18:53.2	2	72	681K	08	3552K	681	1	sleeping	*[112]	0.00000	0.00000	0	7848	145	2498	18544	45822	28484	1312	298	0	0.4			
382	locationd	0.4	00:05:63.7	4	915	9985K	08	3948K	382	1	sleeping	[0]	0.00000	0.38328	205	18257	179	88612	7183	71526	18792	48616	43214	3274	9	0.4		
334	logd	0.4	00:09:55.4	3	2846	22M	08	29M	334	1	sleeping	[0]	0.00000	0.00000	0	28478	98	59378	68588	329278	71573	125414	66	1	0.4			
618	reportd	0.3	00:10:49.3	2	287	19M	08	4528K	618	1	sleeping	[0]	0.00000	0.38525	501	13884	217	4377	4832	36446	11887	17857	1842	1	0.3			
818	netbiosd	0.3	00:01:49.7	6	31	2752K	08	1824K	818	1	sleeping	[0]	0.00000	0.00000	222	5544	184	165	58	92487	719	25555	17	0	0.3			
542	mysqld	0.3	00:18:04.36	8	58	450M	08	448M	542	1	sleeping	[0]	0.00000	0.00000	74	34838	131	1649	888	651126	1481	92874	1651	621	0.3			
749	netatpd	0.3	00:18:57.4	4	94	4546K	08	1252K	749	1	sleeping	[0]	0.00000	0.24956	268	6458	127	6777	1325	38821	6423	13318	476	0	0.3			
684	aid	0.3	00:01:49.5	3	136	7128K	08	4808K	684	1	sleeping	[0]	0.00000	0.24682	501	11373	152	4239	2455	35164	11334	14781	1860	0	0.3			
761	LogiSim-eval	0.2	00:08:58.41	2	374	27M	08	192M	761	1	sleeping	[0]	0.00000	0.00000	501	59773	347	12943	4631	84945	35687	77675	7291	289	0.2			
1813	Google Chrom	0.2	00:18:53.3	2	179	724M	08	179M	641	441	sleeping	[0]	0.00000	0.00000	501	168538	2695	52541	45783	71295	185547	13564	5880	198	0.2			
1183	Comet Helper	0.2	00:08:58.17	1	134	26M	08	21M	965	965	sleeping	[0]	0.00000	0.00000	501	15563	887	13857	5883	2639	12588	7339	48	1	0.2			
376	asnd	0.2	00:07:53.4	3	398	9873K	08	1924K	376	1	sleeping	[0]	0.00000	0.00000	0	36257	161	29347	13494	488362	182013	67688	1183	0	0.2			
1502	Google Chrom	0.2	00:14:47.24	1	248	331M	08	159M	441	441	sleeping	[0]	0.00000	0.00000	501	258469	1958	91624	55485	145107	132785	145624	6189	2	0.1			
482	corebrightness	0.2	00:04:08.2	2	1126	4112K	08	2128K	482	1	sleeping	[0]	0.15756	0.00000	0	8641	129	15149	12779	138273	33349	41791	877	0	0.2			
593	cfprefd	0.1	00:07:76.3	2	876	4112K	33K	1688K	593	1	sleeping	[0]	0.00000	0.15215	501	43562	91	21346	28095	282124	54228	67189	2	0	0.1			
775	spotlightk	0.1	00:18:42.4	2	774	7745K	08	2644K	775	1	sleeping	[0]	0.00000	0.00000	501	12853	178	3864	1983	48826	7614	34853	1154	0	0.1			
363	mds	0.1	00:18:53.8	5	328	33M	08	89M	363	1	sleeping	[0]	0.01277	0.00000	0	57371	158	17882	10287	131988	14884	13851	1268	41	0.1			
512	audiodcheck	0.1	00:08:77.4	3	58	2736K	08	1397K	512	1	sleeping	[0]	0.00000	0.00000	0	1296	89	1572	6851	18835	13415	14778	182	1	0.1			
749	reactivity	0.1	00:18:53.1	1	1854	3652K	08	1854K	749	1	sleeping	[0]	0.00000	0.11258	501	46639	178	3864	1983	48826	7614	34853	1154	0	0.1			
338	fsaved	0.0	00:07:08.12	1	176	4272K	08	1184K	338	1	sleeping	[0]	0.00000	0.00000	0	18916	188	16267	6834	386353	37845	49647	12	54	0.0			
348	powerd	0.0	00:01:96.3	2	153	4992K	08	1368K	348	1	sleeping	[0]	0.00000	0.04699	0	12848	128	18378	18749	46119	29815	22259	269	2	0.0			
1111	Comet Helper	0.0	00:18:53.17	1	122	36M	08	31M	965	965	sleeping	[0]	0.00000	0.00000	501	12853	178	3864	1983	48826	7614	34853	1154	0	0.0			
432	airport	0.0	00:15:24.9	7	367	12M	08	3848K	432	1	sleeping	[11]	0.00000	0.00000	0	33743	172	1401	25748	236118	6784	112968	2268	13	0.0			
1524	PerfPowerd	0.0	00:08:59.7	7	469	12M	08	750K	1524	1	sleeping	[0]	0.00000	0.00000	0	16472	156	65456	32851	56647	34828	5484	1892	3	0.0			
1524	PerfPowerd	0.0	00:08:59.7	7	469	12M	08	750K	1524	1	sleeping	[0]	0.00000	0.00000	0	16472	156	65456	32851	56647	34828	5484	1892	3	0.0			
389	distnoted	0.0	00:08:59.6	2	216	2176K	08	328K	389	1	sleeping	[0]	0.00000	0.00000	0	6723	193	2173	1023	2173	2173	2173	2173					

Command	Resource/Information	OS Concept
uname -a	Kernel and system information	Kernel abstraction
lscpu	CPU architecture and cores	CPU management
ps aux	Running processes	Process management
top	Real-time CPU/memory/process usage	Resource monitoring

Conclusion

The experiment demonstrates that the **operating system acts as an abstraction layer between applications and hardware**. The `uname` command provides information about the Linux kernel and system architecture, while `lscpu` reveals CPU resources managed by the OS. The `ps` command shows how the OS manages individual processes, and `top` provides real-time information about CPU, memory, and process utilization.

The OS abstracts **CPU through process scheduling, memory through virtual memory, storage through files and file systems, and I/O devices through device drivers and system calls**. Therefore, applications can use hardware resources without directly communicating with the physical hardware. This abstraction improves **security, resource sharing, portability, and efficient system management**.